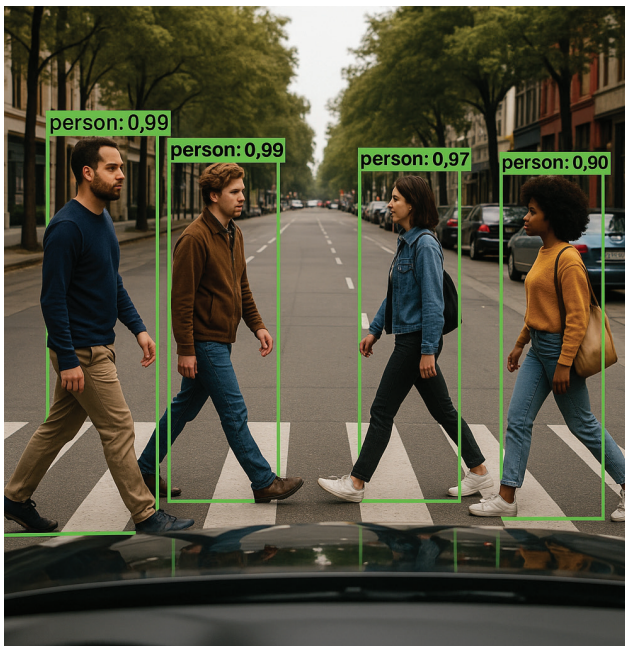


Autonomous Vehicles: Detecting Pedestrians Using YOLOv5

Using AI to identify and track pedestrians in real-time is helping autonomous vehicles become smarter and safer than ever before. Here's an outline of an indigenous project developed to do just that using YOLOv5, a deep learning object detector.



On a cool evening in Bangalore, as I was coming out of my office, I saw an electric shuttle come to a smooth halt as a child dashed across a crosswalk inside our tech park. No driver. No delay. Just action. That's when I realised that autonomous vehicles aren't just futuristic concepts anymore—they're real and learning how to navigate the very same unpredictable streets that we humans do every day when driving our vehicles.

Behind this seamless stop was an intelligent system trained to spot, track, and respond to every human movement. So let's talk about what brought about this vision system's intelligence to life—a system that combines YOLOv5, one of the fastest and best deep learning object detectors, with SORT, a lightweight yet powerful tracking algorithm. Together, these allow machines to detect, recognise and follow pedestrians in each frame as they manoeuvre using the sensor systems they are equipped with.

This isn't just a lab demo or a theoretical model that we are talking about here. This is a real-world implementation

that I have seen and helped to develop. It's been built with excellently engineered deployment in mind—starting from the way it selects the visual clutter to the way it assigns unique IDs to every person who appears in its view.

And, most importantly, this intelligent AI system prioritises human safety.

The challenge of human movement

While vehicles follow lanes and signals, pedestrians generally don't. They hesitate sometimes, may reverse their course, may get distracted by phones, or emerge from occluded spaces like other parked vehicles or some tree lines. Tracking such unpredictable movement requires more than just detection—it needs some kind of memory.

This is where YOLOv5 and SORT excel. YOLOv5 helps to identify all objects present in a scene, and isolates only the 'person' class from these. SORT connects the dots between frames, maintaining a person's identity across both space and time, even when two individuals cross paths or temporarily vanish from view.

Inside the project workflow

You can imagine this system as a digital eye that is connected to a short-term memory.

- YOLOv5 detects every human in the video feed.
- These detections are then filtered to retain only the pedestrian class.
- SORT uses the detections captured to assign each pedestrian with a consistent tracking ID.
- In the end, as an output, a bounding box is drawn around each pedestrian class object in the video, along with persistent labelling.

Each frame thus becomes a story, not just about *who* is there, but also about *where* they're going. The complete architecture of this tracking system is described in Figure 1, highlighting how each module interacts from the moment a video is captured to the point where tracked results are rendered.

The code behind the magic

Here's a peek into the core logic, built in Python using PyTorch and OpenCV: